

Phase change materials (PCMs) for building energy efficiency: A review

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Abstract. Phase change materials (PCM) represent an innovative technology with high potential for increasing the energy efficiency of buildings. These materials can reversibly change their state of matter at a specific temperature, effectively absorbing or releasing a large amount of latent heat during phase changes. Thanks to this property, PCMs can effectively regulate the indoor temperature of buildings, increasing occupant comfort and reducing heating and cooling energy costs. Unlike conventional building materials, they provide temperature stabilization at nearly constant temperatures, which helps to minimize temperature fluctuations in interiors. The paper focuses on the characteristics of PCM materials, the principle of their operation, and their range of applications in building structures to create sustainable and energy-efficient buildings. The aim of the paper is to provide an overview of PCM materials, their potential use in increasing the energy efficiency of buildings, and their contribution to sustainable architecture, including reducing operating costs.

1 Introduction

Phase Change Materials (PCMs) are substances that can change their state of matter at a specific temperature. The most common change is from a solid to a liquid (melting), but it can also occur from a solid to a gas or from a liquid to a gas (boiling). They are capable of releasing, absorbing, and storing a significant amount of energy to provide the necessary heating or cooling during the phase change [1]. When a material changes from a solid to a liquid state or vice versa, its internal structure changes, leading to the absorption or release of heat. Unlike standard materials, when PCMs reach their phase change temperature (melting point), they absorb a large amount of heat at a nearly constant temperature until all the material has melted. As the temperature of the surroundings around the liquid material decreases, the PCM solidifies and releases its stored latent heat [2]. Phase change materials can exist in several different forms, which differ from each other in structure. These forms are called phases. During a phase change, the physical properties of the material change significantly, such as its electrical conductivity, light reflectivity, density, or thermal conductivity [3].

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2 Evolution of PCMs

PCM materials play a crucial role in various industries, including construction, textiles, electronics, and many others, where maintaining a constant temperature is essential. Their ability to store and release heat enables efficient temperature regulation in buildings [4]. Energy absorbed or released by a substance during a phase change (Figure 1.) is known as latent heat [5].

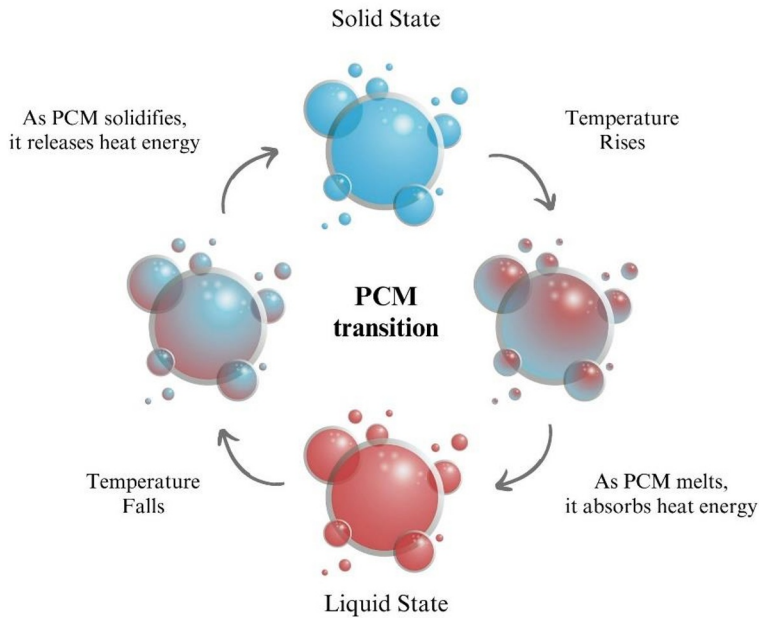


Fig. 1. Phase changes in thermal cycle (according to [6]).

Latent heat is significantly greater than the heat absorbed or released by a material during ordinary temperature changes without a phase transition. In the construction industry, these materials are used to minimize temperature fluctuations indoors, leading to increased occupant comfort and optimized energy costs [7].

The history of PCM materials dates to the latter half of the 20th century. Maria Telkes, a Hungarian American scientist, is credited with pioneering the use of PCM materials as a form of passive heating [8]. Telkes invented a heating system utilizing Glauber's salt for the 'Dover Sun House' in Massachusetts, constructed in 1948. Fans circulated heated air through containers filled with the salt, causing it to melt and store heat at a constant temperature. When the ambient temperature decreased, the salt solidified, releasing the stored thermal energy back into the space [9]. A few years later, researchers focused on organic substances, particularly paraffin, which could undergo phase changes and store heat. In the 1970s and 1980s, the first commercial applications of PCMs appeared in the textile industry to produce thermoregulating garments. The 1990s marked a significant increase in the exploration of PCMs for building applications, aiming to improve energy performance. Alongside these early implementations, extensive research was conducted to investigate the potential of PCMs in construction [10]. Since 2000, the construction industry has shown increasing interest in PCMs, driving the development of commercial products such as building materials and insulation incorporating PCMs. Ongoing research focuses on creating more efficient and sustainable PCM-based solutions to contribute to innovations in building energy efficiency [11].

3 Classification of PCMs

A diverse array of compounds find application as phase change materials, leveraging their unique physical and chemical properties. Solid and liquid PCMs are particularly suitable for building applications, given their capacity to absorb and release substantial amounts of latent heat over a narrow temperature range. The market offers a wide selection of PCMs with tailored melting temperatures [12]. Based on their chemical composition, PCMs can be categorized into organic compounds, inorganic compounds, and eutectic mixtures (Figure 2.). Each category possesses a specific melting temperature range. Salt hydrates, fatty acids, paraffin waxes, and eutectic organic or inorganic compounds have dominated the PCM market for the past three decades [13].

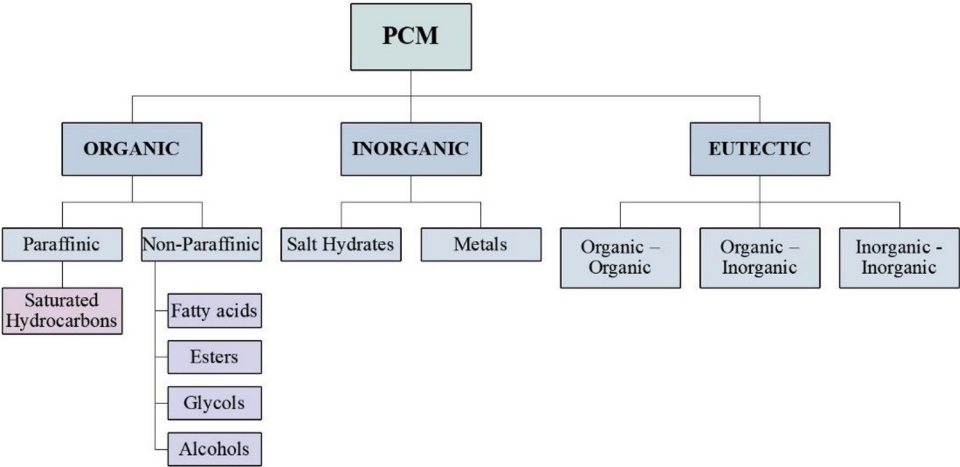


Fig. 2. Classification scheme of PCM materials (according to [14]).

3.1 Composite phase change materials

Composite PCMs constitute a distinct category of PCMs, offering more advanced functionalities. These materials are composed of two or more components, including at least one PCM, and combine the benefits of phase change with the properties of other materials. It allows for tailored applications and improved performance. For example, carbon-based composites demonstrate a substantial increase in thermal conductivity compared to pure PCMs, reaching up to 576% in some cases [15]. A simple example is a copper mesh immersed in paraffin wax. The copper mesh in paraffin wax can be considered a composite material, also known as a thermal composite. Such hybrid materials are created to achieve specific properties. An example is encapsulating paraffin within different silica nanospheres to increase the heat transfer rate [16]. Further development of composite PCM materials brings a new concept of combining PCMs with biomass materials. This innovative approach presents numerous advantages, including reduced environmental impact and the utilization of renewable resources. The use of biomass broadens the application range of composite PCMs and opens up a new direction in their research and development [17].

Multiple approaches can be employed to fabricate composite PCM materials, each offering distinct advantages and the potential for optimization. Composite PCMs can be categorized according to various parameters, such as structural configuration, manufacturing processes, constituent materials, etc. (Table 1.) [18].

Table 1. Categorization of composite PCM materials (according to [19]).

Simple categorization system for composite PCM materials				
Structure	Manufacturing process	PCM type	Matric type	Additive type
Matrix, Encapsulated, Foam.	Impregnation, Polymerization, Encapsulation.	Organic, Inorganic, Eutectic.	Metal, Polymer, ceramic, etc.	To improve thermal conductivity, stability, or fire resistance.

The classification of composite PCM materials is not precisely defined and varies across different literature. There are many types that cannot be definitively categorized into any of these categories [20].

4 Applications in construction

PCM materials applications require specific thermal, physical, chemical, and kinetic properties. The phase change temperature must be suitable for the intended application. Moreover, high thermal conductivity in both, the liquid and solid phases is essential for efficient heat absorption and release [21]. The latent heat of fusion per unit volume must be high to achieve a higher storage density and enable energy storage. A consistent melting behaviour is also significant to ensure reproducible phase change with stable cycling and a long lifetime. Environmental sustainability and economic viability are additional factors that must be considered when selecting PCMs for production [22]. Heat storage plays a vital role in reducing energy consumption. The integration of PCM materials in building construction has become a popular strategy for long-term energy savings and is currently a focus of increased research and development [23]. Due to their high latent heat, stable chemical properties, integrability, and compatibility with building materials, PCMs have been used for heat storage in construction since before 1980. Since then, applications have been updated in various places, such as Trombe walls [24], shutters [25], windowsills [26], and numerous other elements. PCMs are designed to absorb thermal energy during the day and release it gradually at night. They can be directly integrated into building elements like floors, walls, and ceilings. PCM applications in the construction sector are categorized into active and passive systems [27].

4.1 Active systems

Active application systems require mechanical equipment or an additional energy source to operate. These systems are more suitable for cases where higher heat transfer performance or better control is needed. Active heat storage systems with PCM include systems inside the building envelope for heating, ventilation, and air conditioning. Additionally, systems integrated into the building structure and operate through air or water distribution, as well as systems located outside the building envelope, such as domestic hot water storage tanks, etc. [28].

4.2 Passive systems

Passive PCM applications leverage natural energy sources and architectural design to reduce a building's energy consumption. These systems do not require external energy or mechanical equipment, as the charging and discharging of heat is driven solely by temperature variations [29]. Passive applications involve the integration of PCMs into building materials and components. An efficient solution for saving building energy is to embed PCMs into building materials, which apply to the structure's interior or exterior surface. Most often, these are structures such as walls [30], floors [31], ceilings [32], windows, or transparent surfaces [33].

Many researchers are dealing with this issue, and based on their experiments, they have confirmed the effectiveness of embedded PCM materials. The results vary depending on climatic conditions, the location of PCMs in the construction, and the type of material used. For instance, in a study by researchers Nor Ali et al., who integrated PCM into exterior plaster, a decrease in the surface temperature of the wall by up to 9°C was observed [34]. In another study, Salih et al. applied PCM to the inner layer of an external wall and thereby achieved a reduction of the maximum internal room temperature by 4°C [35]. Through simulations, Millers et al. demonstrated the effectiveness of a cooling PCM ceiling panel, achieving a 9 °C reduction in overheating [36].

5 Conclusion

PCMs have a wide range of applications in construction, from insulation and cladding to coatings and plasters. Their use brings many benefits, such as increased thermal comfort, reduced energy consumption of buildings, improved indoor environment, and reduced greenhouse gas emissions. Despite many advantages, there is a need to further research and develop PCM materials to enhance their properties and reduce production costs. The future of PCMs in construction is promising, and their use can significantly contribute to sustainable development and energy efficiency of buildings.

Further development of technologies and processes related to PCM should also focus on their environmental aspect, including recyclability and minimizing the environmental impact of production. Through these steps, PCM material can play a key role in the future of construction, which is increasingly focused on sustainability, energy efficiency, and environmental protection. PCMs represent not only a technological innovation but also a significant step towards greener and more energy-efficient buildings that will meet the demands of the 21st century.

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References

1. C. Liu, C. Wu, Y. Zhao, Z. Chen, T. Ren, Y. Chen, G. Zhang, *Actively and reversibly controlling thermal conductivity in solid materials*, Physics Reports, **1058**, 1-32 (2024)
2. Z. Said, A. K. Pandey, A. K. Tiwari, B. Kalidasan, F. Jamil, A. K. Thakur, V. V. Tyagi, A. Sari, H. M. Ali, *Nano-enhanced phase change materials: Fundamentals and applications*, Progress in Energy and Combustion Science, **104**, 101162 (2024)
3. S. Raoux, *Phase Change Materials*. Annual Review of Materials Research, **39**(1), 25-48 (2009)

4. V. J. Reddy, M. F. Ghazali, S. Kumarasamy, *Innovations in phase change materials for diverse industrial applications: A comprehensive review*, Results in Chemistry, **8**, 101552 (2024)
5. K. Ghasemi, S. Tasnim, S. Mahmud, *PCM, nano/microencapsulation and slurries: A review of fundamentals, categories, fabrication, numerical models and applications*, Sustain. Energy Technol. Assessments, **52**, 102084 (2022)
6. Md. S. Mohtasim, B. K. Das, *Biomimetic and bio-derived composite Phase Change Materials for Thermal Energy Storage applications: A thorough analysis and future research directions*, J. Energy Storage, **84**, 110945 (2024)
7. I. Sarbu, C. Sebarchievici, *A Comprehensive Review of Thermal Energy Storage*, Sustainability, **10**(1), 191 (2018)
8. P. A. Fokaides, A. Kylili, S. A. Kalogirou, *Phase change materials (PCMs) integrated into transparent building elements: a review*, Mater Renew Sustain Energy **4**, 6 (2015)
9. J. M. Charles, *Performance and Stability of CaCl₂·6H₂O-Based Phase Change Materials*, Lehigh University, (2018)
10. M. Chen, L. Jiang, S. Mi, X. Shen, *Energy saving simulation of phase change materials in the enclosure structure of archives warehouse in Lhasa*, Energy Reports, **8**, 4 (2022)
11. H. M. Teamah, *Chapter 1 - Introduction and history of phase change materials' heat transfer*, Phase Change Materials for Heat Transfer, Elsevier, 1-26 (2023)
12. M. Li, Z. Wu, *A review of intercalation composite phase change material: Preparation, structure and properties*, Renew. Sustain. Energy Rev., **16**, 4 (2012)
13. F. Souayfane, F. Fardoun, P. H. Biwole, *Phase change materials (PCM) for cooling applications in buildings: A review*, Energy Build., **129**, 396-431 (2016)
14. K. Ghasemi, S. Tasnim, S. Mahmud, *PCM, nano/microencapsulation and slurries: A review of fundamentals, categories, fabrication, numerical models and applications*, Sustain. Energy Technol. Assessments, **52**, 102084 (2022)
15. P. M. Gilart, Á. Y. Martínez, M. G. Barriuso, C. M. Martínez, *Development of PCM/carbon-based composite materials*, Sol. Energy Mater. Sol. Cells, **107**, 205-211 (2012)
16. G. V. Belessiotis, K. G. Papadokostaki, E. P. Favvas, E. K. Efthimiadou, S. Karellas, *Preparation and investigation of distinct and shape stable paraffin/SiO₂ composite PCM nanospheres*, Energy Convers. Manage., **168**, 382-394 (2018)
17. Q. Zhang, J. Liu, J. Zhang, L. Lin, J. Shi, *A Review of Composite Phase Change Materials Based on Biomass Materials*, Polymers, **14**(19), 4089 (2022)
18. S. R. L. da Cunha, J. L. B. de Aguiar, *Phase change materials and energy efficiency of buildings: A review of knowledge*, J. Energy Storage, **27**, 101083 (2020)
19. X. Du, H. Wang, Z. Du, X. Cheng, *Synthesis and thermal energy storage properties of a solid–solid phase change material with a novel comb-polyurethane block copolymer structure*, Thermochimica Acta, **651**, 58-64 (2017)
20. Y. E. Milián, A. Gutiérrez, M. Grágeda, S. Ushak, *A review on encapsulation techniques for inorganic phase change materials and the influence on their thermophysical properties*, Renew. Sustain. Energy Rev., **73**, 983-999 (2017)
21. H. Togun, H. S. Sultan, H. I. Mohammed, A. M. Sadeq, N. Biswas, H. A. Hasan, R. Z. Homod, A. H. Abdulkadhim, Z. M. Yaseen, P. Talebizadehsardari, *A critical review on phase change materials (PCM) based heat exchanger: Different hybrid techniques for the enhancement*, J. Energy Storage, **79**, 109840 (2024)

22. K. Y. Leong, M. R. A. Rahman, B.A. Gurunathan, *Nano-enhanced phase change materials: A review of thermo-physical properties, applications and challenges*, J. Energy Storage, **21**, 18-31 (2019)
23. G. Simonsen, R. Ravotti, P. O'Neill, A. Stamatiou, *Biobased phase change materials in energy storage and thermal management technologies*, Renew. Sustain. Energy Rev., **184**, 113546 (2023)
24. Y. Zhang, Z. Zhu, Z. Peng, J. Luo, X. Sun, J. Li, F. Peng, *The Trombe wall equipped with PCMs for the enhancement of the indoor thermal quality*, Energy Build., **297**, 113407 (2023)
25. M. Che-Pan, E. Simá, A. Ávila-Hernández, J. Uriarte-Flores, R. Vargas-López, *Thermal performance of a window shutter with a phase change material as a passive system for buildings in warm and cold climates of México*, Energy Build., **281**, 112775 (2023)
26. M. Musiał, L. Lichołai, *Multi-Faceted Analysis of Phase-Change Composite Intended for Autonomous Buildings*, Materials, **17**, 2604 (2024)
27. M. Ghamari, C. H. See, D. Hughes, T. Mallick, K. S. Reddy, K. Patchigolla, S. Sundaram, *Advancing sustainable building through passive cooling with phase change materials, a comprehensive literature review*, Energy Build., **312**, 114164 (2024)
28. G. Gholamibozanjani, M. Farid, *A comparison between passive and active PCM systems applied to buildings*, Renewable Energy, **162**, 112-123 (2020)
29. K. Jiao, L. Lu, L. Zhao, G. Wang, *Towards Passive Building Thermal Regulation: A State-of-the-Art Review on Recent Progress of PCM-Integrated Building Envelopes*, Sustainability, **16**(15), 6482 (2024)
30. P. Zhang, X. Xiao, Z. W. Ma, *A review of the composite phase change materials: Fabrication, characterization, mathematical modeling and application to performance enhancement*, Applied Energy, **165**, 472-510 (2016)
31. W. Sun, Y. Zhang, Z. Ling, X. Fang, Z. Zhang, *Experimental investigation on the thermal performance of double-layer PCM radiant floor system containing two types of inorganic composite PCMs*, Energy Build., **211**, 109806 (2020)
32. H. Lim, Y. K. Kang, J. W. Jeong, *Application of a phase change material to a thermoelectric ceiling radiant cooling panel as a heat storage layer*, J. Build. Eng., **32**, 101787 (2020)
33. C. Liu, Y. Wu, Y. Zhu, D. Li, L. Ma, *Experimental investigation of optical and thermal performance of a PCM-glazed unit for building applications*, Energy Build., **158**, 794–800 (2018)
34. U. N. Nor Ali, N. M. Nor, N. A. Misnon, S. A. M. Noor, M. Othman, M. A. A. Alias, A. Syamsir, *Integration of PCM as an external wall layer in reducing excessive heat of building walls*, Funct. Compos. Struct., **4**(3), 035005 (2022)
35. S. S. O. Salih, I. I. A. Hamakhan, A. A. Abdulkader, *Investigation of Thermal Performance of 3D Printing Integrated Phase Change Materials in Building Structure*, International Journal of Heat and Technology, **40**(3), 843-848 (2022)
36. R. Millers, A. Korjamins, A. Lešinskis, A. Borodinecs, *Cooling Panel with Integrated PCM Layer: A Verified Simulation Study*, Energies, **13**(21), 5715 (2020)